

MANDEEP KHATRI

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EDUCATION

Alabama A&M University

Expected Graduation: December 2027

Bachelor of Science in Computer Science, **GPA: 3.80/4.0**

Huntsville, AL

- Awards & Honors:** Dean's List, AAMU Merit Full Scholarship, EICOP 2025 Finalist
- Relevant Coursework:** Data Structures & Algorithms, Intro to programming I and II (Python), Computer Architecture, Software Engineering, (Operating Systems – Planned), Discrete Mathematics, Linear Algebra

TECHNICAL SKILLS

Languages: Python, C/C++, MATLAB, TypeScript/JavaScript, Go, SQL, Bash, HTML, CSS

Frameworks & Tools: Flask, React, FastAPI, Angular, Node.js, Streamlit, FAISS, Django, PyMuPDF, Docker, Linux, Agile/Scrum, SQLite, REST APIs, MQTT, PyTest, JMeter, GraphQL

AI/ML & Cloud: AWS (Lambda, EC2, S3), GCP (Vertex AI, Gemini API), TensorFlow, PyTorch, Hugging Face, LangChain, RAG, MLOps (MLflow, Kubeflow, Kubernetes)

Build/Tooling: Git, GitHub Actions (CI), Jest/Playwright, Vite/Webpack

CERTIFICATIONS

Google Cloud Certified – Associate Cloud Engineer

July 2025

WORK EXPERIENCE

Academic Teaching Assistant

January 2025 – Present

Alabama A&M University

Huntsville, AL

- Assist instruction for **80+ freshmen** in **C++, Python, and Calculus I**; prepared course materials, guided labs, and graded assignments, contributing to a **15%** improvement in exam performance.
- Conduct **LeetCode-style** workshops, **live coding demos**, & project-based labs using GitHub Classroom, & **algorithm visualization tools**, mentoring students in collaborative coding and technical problem-solving.

Undergraduate Researcher – Autonomous Systems Lab

August 2024 – Present

Alabama A&M University

Huntsville, AL

- Developed a **GPU-accelerated vision system** for autonomous drones using **MATLAB, C++, and CUDA**, boosting real-time image-processing speed by **3×** through optimized parallel computation.
- Trained a **YOLOv5 deep learning model** with **MATLAB's Deep Learning Toolbox** and **Simulink**, achieving **92% obstacle-detection accuracy** in simulated flight environments.

Cloud Computing Intern

January 2023 – March 2023

Verisk

Kathmandu, Nepal

- Built a cloud-native IoT monitoring system with **Python & C++**, deploying microservices on **AWS EC2** & integrating MQTT protocol for device communication; boosted real-time data ingestion & processing efficiency by **30%**.
- Developed a responsive analytics dashboard with **HTML, CSS, & JavaScript**, backed by **Flask/Express RESTful APIs** for live sensor data; reduced user query and visualization time by **20%**.
- Conducted profiling and load testing with **PyTest, JMeter, & Dockerized** containers, resolving bottlenecks & lowering system latency by **25%**, while improving fault tolerance & aligning development with **Agile/SDLC** practices.

PROJECTS

Memory Tiering Simulator | C++, Python, Linux, Benchmarking

January 2025 – Present

- Simulated a **two-tier memory hierarchy** (fast DRAM + slow tier) and implemented **hot-page detection** with **auto-migration** in **C++**, improving average access latency by **30%**.
- Benchmarked performance using **YCSB workloads** and visualized **tier utilization** with **Python** to analyze **throughput** and **memory efficiency**.

LLM-Powered Course Q&A Chatbot | Python, Gemini API, LangChain, Streamlit

April 2024 – Present

- Built a Q&A chatbot using **Python, Gemini 1.5 Pro API (via Vertex AI)**, and **LangChain**, processing **100+** pages of course PDFs with **PyMuPDF**, text splitting, and embedding via **Gemini**; reached **85%** manual QA accuracy.
- Implemented **RAG** pipeline with **FAISS** (dense retrieval) and **Streamlit** UI, optimizing prompt templates and reducing avg. response latency by **40%** vs. keyword search baseline.

ACTIVITIES

Google Cloud Career Jumpstart HBCU Fellow

May 2025 – August 2025

HBCU Possibilities Summit

January 2025 – May 2025

Apprenticeship Universe Program - HBCU 20x20

May 2025

TracerFire Cybersecurity Competition - Sandia National Laboratories

September 2024